

The Volatility Targeted Coupled Factor Portfolio

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Abstract

This document describes a volatility targeted coupled factor portfolio for monthly construction. A network maps each security’s characteristics to a set of latent factor loadings conditioned on the cross section, a second network estimates each security’s idiosyncratic variance and supplies an inverse variance weighting, the resulting factor portfolios are combined by a mean variance rule estimated from their own history, and the combined book is neutralised to market beta and scaled to a volatility target. We present each stage in turn and connect the design to the evidence on selection skill and risk timing.

1 Overview and motivation

Rather than mapping characteristics to a single score, the model maps them to the loadings of a small number of latent long short factors, treats those factors as the primitive tradeable objects, and then combines them by a portfolio rule. This two stage construction separates the question of what the factors are, which is answered by the loading network, from the question of how they are combined, which is answered by a mean variance rule on their realised returns. The separation allows the risk model to enter twice, once through an inverse variance weighting of securities within a factor and once through the covariance of the factors in their combination, while the leverage of the final book is controlled by an explicit volatility target.

2 The shared representation

The raw inputs are a panel of monthly characteristics and a list of candidate features, and the target is the excess return of a security over the following month. In each month every retained characteristic is transformed to its centred within month percentile rank, missing entries are set to the median before centring, and securities missing more than one third of the characteristics are dropped. The coverage filter that selects the characteristics keeps a characteristic if it is present in at least ninety percent of the months at or before the cutoff, and it is recomputed each year. The networks are refitted yearly on the months whose forward returns precede the year, with warm starting from the previous year and averaging over four seeds.

3 The loading network and the factor portfolios

The first network maps the characteristic vector of each security to a vector of $k = 32$ latent loadings, and it is conditioned on the cross section through a pooled context, that is the mean of the security embeddings, so that a loading depends on the composition of the month rather than on the security alone. Writing $H \in \mathbb{R}^{n \times k}$ for the matrix of loadings, the columns are demeaned across

securities. Each column is then turned into a long short factor portfolio of unit gross exposure,

$$W_{\cdot c} = \frac{\tilde{H}_{\cdot c}}{\sum_i |\tilde{H}_{ic}|}, \quad \tilde{H}_{\cdot c} = H_{\cdot c} - \overline{H_{\cdot c}}, \quad (1)$$

so that $W \in \mathbb{R}^{n \times k}$ collects k portfolios and the realised return of factor c in a month is $f_c = \sum_i W_{ic} r_i$, that is $f = W^\top r$.

4 The precision network and inverse variance weighting

The second network supplies the security level risk weighting. It maps the characteristics of a security to the logarithm of its idiosyncratic variance, and it is fitted by Gaussian likelihood on the residuals of the factor model. For a training month the factor returns are first recovered by a ridge regression of returns on loadings,

$$\hat{f} = (H^\top H + \eta I)^{-1} H^\top r, \quad (2)$$

and the squared residual of security i is $e_i^2 = (H\hat{f} - r)_i^2$. Writing ℓ_i for the predicted log variance, the network minimises the Gaussian negative log likelihood

$$\mathcal{L} = \frac{1}{n} \sum_i \left(\ell_i + e_i^2 e^{-\ell_i} \right). \quad (3)$$

The predicted precision is used to weight each security's contribution to the factor portfolios through $\omega_i = e^{-\ell_i}$, normalised to have unit mean across the month, so that securities with lower idiosyncratic variance receive more weight in the formation of the factors.

5 Fitting the loadings on realised Sharpe

Over a batch of training months the loading network is fitted to maximise the realised Sharpe ratio of the equally combined factors. For each month in the batch the equal combination has return $\bar{f} = k^{-1} \sum_c f_c$, and the batch objective is

$$\max \frac{\text{mean}(\bar{f})}{\text{std}(\bar{f})}, \quad (4)$$

where the mean and standard deviation are taken over the months of the batch. The loadings are therefore chosen so that the factors, combined equally, form a portfolio with a high Sharpe ratio, and the precision weighting is applied when the factors are formed.

6 The mean variance combination

The factors are combined for trading not by the equal weighting used in fitting, but by a mean variance rule estimated from their own realised returns. From the history of factor returns we form exponentially weighted moments with a half life of one hundred and twenty months,

$$\hat{\mu} = \sum_t \pi_t f_t, \quad \hat{\Sigma} = \sum_t \pi_t (f_t - \hat{\mu})(f_t - \hat{\mu})^\top, \quad (5)$$

where π_t are the normalised exponential weights. The covariance is shrunk towards a scaled identity and lightly ridged, and the combination is the solution of the mean variance problem in closed form,

$$\lambda = \hat{\Sigma}^{-1} \hat{\mu}. \quad (6)$$

The traded book before overlays is $w = W\lambda$, normalised to unit gross exposure and averaged across the four seeds.

7 Beta neutralisation and volatility targeting

Two overlays are applied to the combined book. First, the component of the book along market beta is projected out without an intercept, so that the residual market beta is removed while the net exposure is preserved,

$$w^\perp = w - b \frac{b^\top w}{b^\top b}. \quad (7)$$

Second, the book is scaled so that its ex ante volatility meets a fixed annual target of ten percent. The ex ante volatility is measured against a Ledoit and Wolf shrinkage covariance estimated from the trailing year of daily returns, and the scaling is

$$w \leftarrow w \cdot \min\left(\frac{v}{\sqrt{w^\top \Sigma w}}, G\right), \quad (8)$$

where v is the monthly volatility target and G is a gross exposure cap that in practice does not bind. Because the Sharpe ratio is invariant to scale, the volatility target changes the time path of leverage but not the underlying selection, and the evidence on this panel records that this overlay alone raised the test Sharpe ratio from 3.123 to 3.442 while leaving the weights unchanged.

8 Evaluation and concluding remarks

On the evaluation window the model attains a Sharpe ratio of 3.324 in the volatility scaled form and 3.447 in the unscaled form. A variance decomposition of the book against eight standard characteristic factors found an annualised alpha of 5.74 percent with a t statistic of 13.1, so that under the null hypothesis of zero alpha we reject at any conventional level and conclude that the book carries statistically significant selection skill. Only 20.6 percent of the book's variance is explained by the common factors, and the information ratio relative to them, at 2.42, is below the realised Sharpe ratio of 2.61 on the unscaled beta neutral book, which suggests that the systematic exposures of the portfolio are, on net, rewarded rather than detrimental. We conclude that the coupled factor construction combines a strong and largely idiosyncratic selection signal with an aggregate risk overlay that times leverage effectively, and that it is the timing of leverage rather than any refinement of security level risk that lifts the Sharpe ratio to its high value.